A-A-54309
31 December 1990

COMMERCIAL ITEM DESCRIPTION

BANDAGE, CAST, POLYESTER FABRIC, RESIN IMPREGNATED

The General Services Administration has authorized the use of the Commercial Item Description.

This commercial item description covers lightweight, rolled, resin impregnated bandages suitable for forming orthopaedic casts.

Salient Characteristics:

Material. Materials used in the manufacture of the bandages shall be of high quality and suitable in all respects for the purpose intended. The substrate fabric shall be a raschel knitted polyester backing of warp chain loops and straight filling threads. The impregnating material used in the manufacture of the bandage shall be a water activated urethane prepolymer resin composition. The resin composition shall cure upon exposure to water to provide a hardened cast bandage.

Construction. The bandages shall consist of a raschel knitted polyester fabric impregnated with the specified resin. When activated by water the bandages shall be moldable to produce a radiolucent hard coated cast with a non-dusting surface. The resin shall be uniformly spread and firmly bonded to the fabric. The finished bandages, when used in accordance with the manufacturer's instructions, shall be suitable for forming orthopedic casts for immobilization purposes. The finished bandages shall appear white (with a tint due to the urethane resin) and shall be furnished as a neatly rolled bandage. The long edges of the rolled bandages shall be parallel with a selvage edge, free from ravelled threads.

The rolled bandages shall be furnished as a roll, in one continuous length of material. Bandages shall be nominally 4 yards (3.66 meters) long and 5 inches (12.7 cm) wide. Each bandage shall be evenly rolled on a rigid non-absorbent core having an inside diameter of 1/2 inch. The moisture sensitive rolled bandages shall be individually protected in a heat-sealable laminated packaging material.

AMSC N/A FSC 6510

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The laminated material shall consist of 2 layers of aluminum foil sandwiched between an inner layer of heat-sealable plastic polymer film and an outer layer of plastic polymer film which protects the aluminum foil from punctures, abrasion and flexural fatigue cracks.

Dimensions. Bandages shall be not less than 5.0 inches wide and 144.0 inches long.

Cast strength. The cast strength shall be not less than 50 lbs. after 15 minutes, not less than 80 lbs. after 1 hour, and not less than 100 lbs. after 24 hours, when tested as specified below.

Water penetration. Bandages shall show uniform and complete water penetration, with no visible dry spots or areas, when tested for cast strength.

Exotherm. The peak exotherm temperature shall not exceed 118°F when the bandages are tested as specified below.

Setting time. The setting time of the bandages shall be less than 8 minutes. At such time the cast shall be rigid enough to immobilize the affected limb.

Working time. The bandages shall have a minimum of 60 seconds working or application time from the time they are immersed in the dip water.

Package integrity. The laminated pouch containing the finished bandage shall not leak when tested as specified below.

Irritation. The bandages shall be non-irritating and shall not cause thermal burns to the normal skin when used as directed. The bandages shall be free from harmful materials.

Instructions for use. Instructions for use shall be furnished for the bandages and shall be included in each individual package of 10 bandages. The printing shall show no offsetting, smearing or bleeding and shall be clearly legible.

Shelf life. The finished packaged bandages shall comply with the requirements of this specification for not less than 24 months from date of manufacture when stored at/or below 75°F (24°C).

Lot or batch number. Each foil laminated pouch shall be marked with a lot or batch number and date of manufacture.

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Delivery. Not more than 4 months shall have elapsed from the date of manufacture of the bandages to the date of delivery to the Government.

Workmanship. Bandages and their enclosing laminated foil pouches shall be free of defects which may detract from their appearance or impair their serviceability. Bandages shall be furnished clean.

Unit. Package (PG). One package containing 30 rolled bandages, as specified, constitutes one unit.

Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of two years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Sampling for inspection. Sampling for inspection shall be conducted in accordance with MIL-STD-105. Where feasible, the same sample shall be used for the determination of two or more inspection characteristics. The unit of product for inspection purposes shall be one package of bandages. Unless otherwise specified, sampling for examination shall be conducted at inspection level II, AQL 2.5, and sampling for tests (when applicable) shall be conducted at inspection level S-1, AQL 1.0.

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Examination. Bandages shall be examined to verify that construction and workmanship are in accordance with applicable requirements.

Tests. Tests shall be conducted to determine compliance with specification requirements.

Dimensions. Length and width of bandages shall be measured.

Standard testing conditions. All testing specified below shall be conducted in a controlled environment with the temperature at 75°F ± 3°F and a relative humidity at 55% ± 5% on bandages wrapped and ready for shipment by removing envelope and testing as specified. Equipment and material shall be conditioned at this temperature and humidity at least ten hours prior to testing.

Cast strength. The test is used to determine the strength of a 2-3/4 inch inside diameter test cylinder when compressed diametrically at a specified test time. Test times are designated at 15 minutes; 1 hour; and 24 hours after activation of the bandage by immersion in water. Samples are removed from the forming dowl at 10 minutes and stored in an open area at ambient conditions until test time.

While wearing protective gloves of vinyl or latex, the tester shall remove the bandage from the pouch just prior to wrapping and dip it into clean, fresh water at 75°F ± 1°. The bandage is squeezed 4 to 5 times to complete water penetration, removed and shaken, then wrapped onto the solid metal forming dowl which has been covered with 1 layer of waxed paper. The complete bandage, approximately 16 layers, is rolled onto the dowl so that the long length edges lay on top of each other. The cylinder is compressed by hand to assure good lamination between layers.

Ten minutes after immersion, the samples are removed from the dowl and stored. At the specified test times the sample cylinders are compressed in an Instron Model 1122 or a Chattelon Tester Model USTM.

The cylinders are placed between the platens of the test apparatus so that the lengthwise direction of the cylinder aligns with the lengthwise direction of both platens and does not exceed the length of the platens on either side. The gap between the platens is closed so that both are in contact with the cylinder but no load is applied. Compression is then begun at a rate of 12 inches per minute for a distance of 1 cm at which point the strength of the cylinder is recorded. This load is recorded in pounds and 4 test results are recorded per lot for each specified time.

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Exotherm. This test determines the peak exothermic heat of a single bandage when wrapped continuously around a 2-3/4 inch wooden dowl. The dowl is grooved lengthwise and has 4 thermocouples spaced evenly around it. The thermocouples and dowl are covered by one layer of waxed paper. The temperature is measured and recorded on a Dynamaster recorder or similar device. The initiation of the bandage reaction is done by dipping it into fresh, clean water at 75°F ± 1° and squeezing 4 to 5 times. The bandage is removed and shaken and rolled onto the dowl so that the long length edges lay on top of each other. The cylinder is compressed by hand to assure good lamination between layers. The peak temperature reached is recorded as the exotherm.

Setting time. This test is done on samples that have been wrapped for strength determination. It is designed to measure the time frame when an individual bandage becomes hard. After completion of wrapping a bandage on the metal dowl for the crush strength samples, the bandage is indented by a fingernail. When the wrapper can no longer make an impression in the surface of the cylinder, using moderate pressure, the sample is set and the time recorded.

Working time. This test is done on samples that are wrapped for cast strength determination. It is designed to measure a minimum amount of time available to apply a single bandage. During the wrapping of a bandage around the metal forming dowl, the time available to unwind each sample is measured from the moment it is dipped into the water until it is completely on the dowl. This time shall be not less than 60 seconds and there should be no undue effort needed to unroll at that time.

Packaging seal integrity. The individually sealed pouch containing a bandage is placed in a water filled vacuum tester. The water is colored with a red dye and the level of the water shall be such that the pouch is completely submerged. The tester is closed and a vacuum equal to 18 inches of mercury is drawn for 1 minute. The submerged pouch is examined for leaks as indicated by bubbles escaping from it. If air bubbles are detected the pouch is rejected.

Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

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Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch/pound units, providing they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this document are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

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NOTE: The following National Stock Number is covered by this document:

National Stock Number	Item Identification
6510-01-308-5343	BANDAGE, CAST, POLYURETHANE RESIN IMPREGNATED, Fast Setting, Rolled, Radiolucent, 5 inches (12.7 cm) Wide, 4 yards (3.66 meters) Long, 30's

Ordering data. Intermediate/exterior package quantities, labeling and marking must be as specified in the contract and/or order.

MILITARY INTERESTS:

Custodians:

Army - MD
Navy - MS
Air Force - 03

PREPARING ACTIVITY:

DoD - MB

Agent:

DLA - DM

CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
PHS
FDA-MPQAS

Project No. 6510-0795
Location:WORD/BANDAGE/RESIN

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
A-A-54309	BANDAGE, CAST, POLYESTER, RESIN IMPREGNATED

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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